

Fully automated electrodeposition of perovskite thin films for efficient solar cells

Cite as: Appl. Phys. Lett. **128**, 263902 (2026); doi: [10.1063/5.0338967](https://doi.org/10.1063/5.0338967)

Submitted: 15 April 2026 · Accepted: 11 June 2026 ·

Published Online: 29 June 2026



View Online



Export Citation



CrossMark

Tao Yu, Yuchong Chen, Zhan Chen, Chenchao Feng, Diyun Xue, Centao Zhu, Chunhe Li, and Kuankuan Ren ^{a)}

AFFILIATIONS

Zhejiang Engineering Research Center of MEMS, School of Mathematical Information, Shaoxing University, Shaoxing 312000, China

^{a)}Author to whom correspondence should be addressed: renkk1991@semi.ac.cn

ABSTRACT

Electrodeposition technology has shown unique application potential in perovskite solar cell fabrication due to its characteristics of requiring only water and alcohol solvents without the need for high-vacuum environments. However, current related research is still in the preliminary exploration stage (fewer than 50 publications), and most are concentrated on manual laboratory preparation, with prominent issues such as rough process control and poor reproducibility, with few systematic studies on automated control of electrodeposition processes and synergistic regulation of additives. Addressing these issues, this work designs and constructs a fully automated electrodeposition preparation system, which achieves high reproducibility in the preparation of MAPbI₃ perovskite solar cells. The resulting devices exhibit an efficiency of 7.29%, a 16% improvement over manual preparation, with the efficiency standard deviation reduced from 1.47% to 1.22%. On this basis, by further introducing MACl and FABr additives, the champion efficiency was improved to 15.15%, with the efficiency standard deviation reduced to 0.93%. The automated preparation system and additive regulation strategy proposed in this study provide effective technical support and equipment reference for developing low-cost and environmentally friendly preparation of perovskite solar cells.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0338967>

In recent years, the laboratory power conversion efficiency (PCE) of perovskite solar cells (PSCs) has reached 27.3%.¹ Their high efficiency, solution processability, and mechanical flexibility demonstrate broad prospects for practical applications.² As the core light-absorbing layer of the solar cells, the perovskite film's microscopic morphology, crystallization quality, and defect states directly determine the device performance and long-term stability. To date, diverse fabrication routes have been developed, including spin coating,^{3–5} blade coating,^{6,7} inkjet printing,^{8–10} screen printing,¹¹ evaporation,^{12,13} and chemical vapor deposition.¹⁴ However, these methods rely heavily on toxic solvents such as dimethylformamide, γ -butyrolactone, chlorobenzene, and toluene, raising environmental and safety risks. Many also require high-vacuum or high-temperature conditions, increasing equipment cost and energy consumption. Accordingly, exploring environmentally friendly, low-cost, and scalable deposition strategies has become an urgent demand for PSCs.

Electrodeposition technology has been applied to the preparation of PSCs due to its characteristics of cost-effectiveness, environmental friendliness, and suitability for ambient temperature and pressure processing.^{15,16} For example, Cui *et al.* used an

electrodeposited PbO combined with a two-step solution immersion to prepare perovskite films, and the PSCs achieved an efficiency of 9%.¹⁷ In 2015, Chen *et al.* used electrodeposited PbO₂ to prepare PSCs with an active area of 4 cm², achieving a PCE of 4.5%, confirming the application potential of electrodeposition technology in large-area PSCs.¹⁸ Lv *et al.* used electrodeposited PbO₂ films as a lead source and successfully prepared all-inorganic perovskite CsPbI_{2–x}Br_{1+x} PSCs with an efficiency of 6.87%.¹⁹ Li *et al.* used MABr-doping FAI to react with electrodeposited PbO₂ and effectively suppressed the formation of δ -phase FAPbI₃, preparing electron transport layer-free PSCs with a PCE of 10.7%.²⁰ In 2025, Xiang *et al.* improved the efficiency of electrodeposited FAPbI₃ PSCs to 19.3% through synergistic optimization of composition, solvent, and passivation engineering.²¹ Xue *et al.* further improved the PCE of electrodeposited PSCs to 21.21% and significantly enhanced device stability by using octylammonium iodide for surface modification of electrodeposition-converted PbI₂.²² Despite these achievements in electrodeposition of PSCs, a fully closed-loop automated electrodeposition system covering the entire perovskite fabrication process has not been reported, leading to problems such as rough process control, poor reproducibility, and large operational errors. Moreover, the

synergistic effect of MAI and FABr in optimizing the performance of electrodeposited PSCs remains unexplored.

Addressing these issues, this study developed a fully automated preparation system for PSCs based on electrodeposition processes. The MAPbI₃ PSCs prepared based on this automated preparation system achieved a PCE of 7.29%, a 16% improvement over manual preparation, with efficiency standard deviation reduced from 1.47% to 1.22%. Meanwhile, we further improved the champion device efficiency to 15.15% by introducing MAI and FABr additives, with the efficiency standard deviation reduced to 0.93%. These results indicate that through the synergistic effect of the automated electrodeposition preparation system and additives, the efficiency and reproducibility of PSCs' preparation were improved, providing important technical support and equipment foundation for developing low-cost and environmentally friendly preparation of PSCs.

The fully automated preparation system designed and constructed in this study is shown in Fig. 1(a). The detailed composition and main performance of the automated preparation system can be found in Table S1. This system mainly consists of a base platform, master control, stepper motor, linear rail, robotic gripper, conductive gripper, power supply, reaction chamber, and gas blowing components connected to the master control, respectively. The robotic gripper can handle substrates weighing 0.01–2 kg with a moving speed of 0.1–3 m/min, while the conductive gripper is made of nickel-plated copper and exhibits a low contact resistance of about 0.015 Ω/cm^2 . The schematic of the automated preparation control system is shown in Fig. S1, where the master control module is used for centralized management of the motor drive module, power control module, and gas blowing control module. Among them, the motor drive module is electrically connected to the motor and drive components, controlling

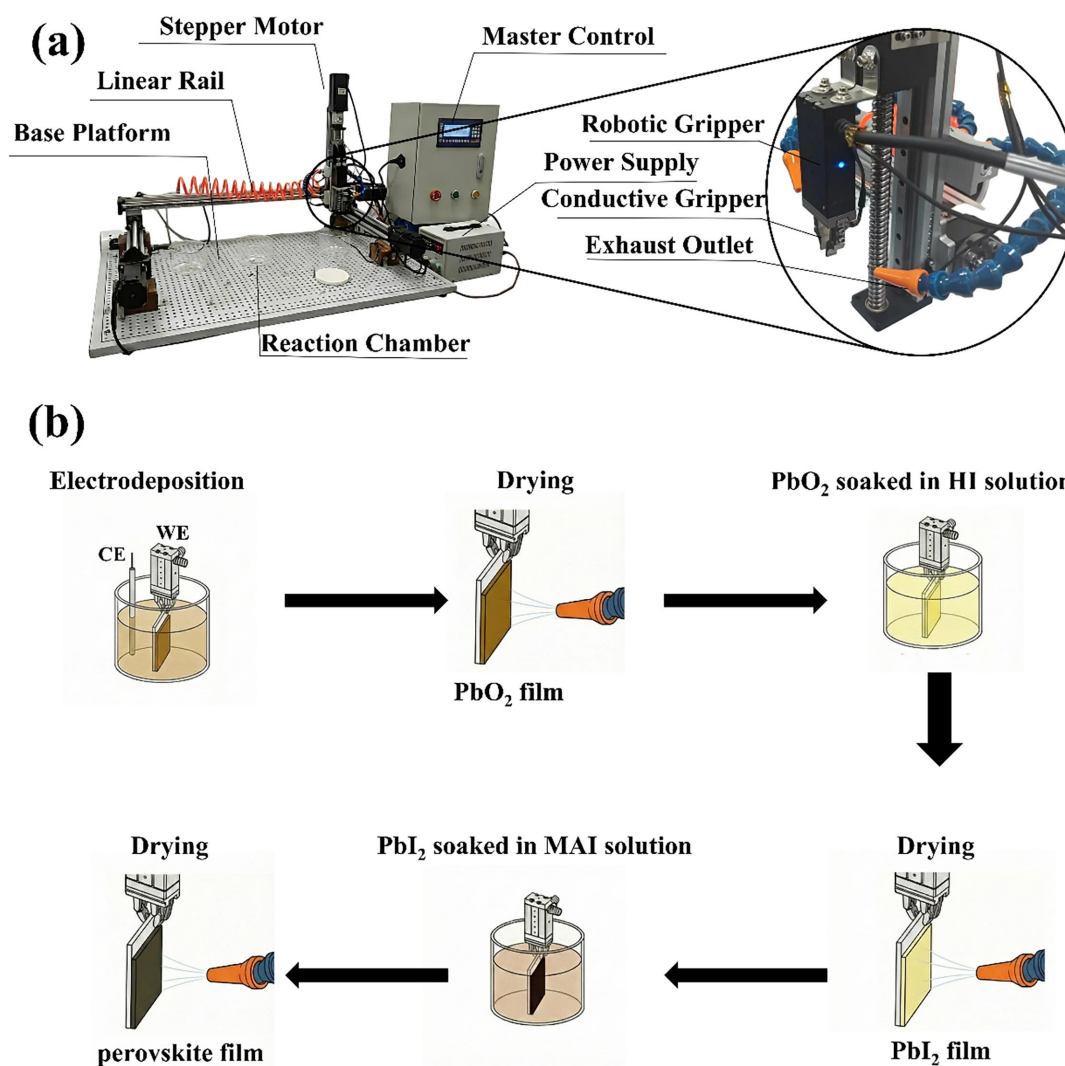


FIG. 1. (a) The image of the automated preparation system for PSCs. (b) Schematic diagram of the preparation process for PSCs based on the automated preparation system.

the movement of the motion stage by regulating motor rotation, and achieving gripper opening and closing by controlling the gripper drive components; the power control module is electrically connected to the power switch for controlling the power supply component on/off; and the gas blowing control module is electrically connected to the exhaust outlet for controlling the gas blowing operation. Through coordinated control of the above modules, this automated system can produce up to 20 samples/h, with a total cost of approximately \$730.

Figure 1(b) shows the process flow schematic of this system for preparing PSCs. When the robotic gripper delivers the ITO/SnO₂ substrate to the reaction chamber and immerses it in the electrolyte, the control module starts the power supply, forming a closed current loop consisting of the power supply anode, mechanical gripper, sample substrate, electrolyte, reaction chamber, and power supply cathode. The control system can precisely set deposition voltage, current, and deposition time, thereby achieving automation and high-precision control of the preparation process. The main process for preparing PSCs based on this system includes the following three consecutive steps. The first step is to directly electrodeposit PbO₂ onto the ITO/SnO₂ substrate, with the reaction equation as follows:



After PbO₂ deposition is completed, it is immersed in an aqueous solution to wash away excess impurities and dried, then immersed in a hydriodic acid (HI)/ethanol solution to convert to PbI₂. This reaction involves Pb⁴⁺ reduction to Pb²⁺ and formation of PbI₂, with the reaction equation as follows:



The obtained PbI₂ is immersed in an ethanol solution to wash away excess impurities and dried, then immersed in a methylammonium iodide (MAI)/isopropanol solution to obtain perovskite MAPbI₃ films, with the reaction equation as follows:



For preparing perovskite films with MAI and FABr additives, only the above MAI/isopropanol solution needs to be replaced with an MAI/isopropanol solution containing MAI and FABr additives, while reaction steps and reaction processes remain unchanged.

To verify the effectiveness of the preparation based on this automated preparation system, the performance of the PSCs prepared by the automated and manual methods was compared. The PSCs adopt an ITO/SnO₂/MAPbI₃/Spiro-OMeTAD/Au structure (Fig. S2), and a photograph of a representative device is shown in Fig. S3. The distribution of PSCs' parameters is shown in Figs. 2(a)–2(d), and the J–V curves of the best PSCs are shown in Fig. 2(e), with detailed photovoltaic parameters summarized in Table S2. From Figs. 2(a)–2(e) and Table S2, it can be seen that PSCs prepared using this automated preparation system consistently outperform manually prepared devices in key performance parameters such as average PCE, average fill factor (FF), average open-circuit voltage (V_{oc}), and average short-circuit current density (J_{sc}). Specifically, the highest PCE of automated prepared MAPbI₃ PSCs reached 7.29%, while manually prepared devices achieved only 6.29%, representing a relative improvement of about 16%. In terms of reproducibility, automated prepared devices had an average PCE of 5.41% with a standard deviation of 1.22%; manually prepared devices had an average PCE of 4.16% with a standard deviation of 1.47%. These results arise from the lack of precise control over the gas-blowing angle, flow rate, and drying uniformity in manual preparation, which leads to non-uniform solvent evaporation, surface defects, and pinholes. In contrast, the automated system precisely controls these processes, ensuring uniform solvent evaporation and reducing trap density and recombination losses, thereby directly improving the photovoltaic parameters. The x-ray diffraction (XRD) patterns of perovskite MAPbI₃ films prepared by automated and manual methods are shown in Fig. 2(f). Both films exhibited characteristic diffraction peaks at 2θ = 14.1° and 28.5°, corresponding

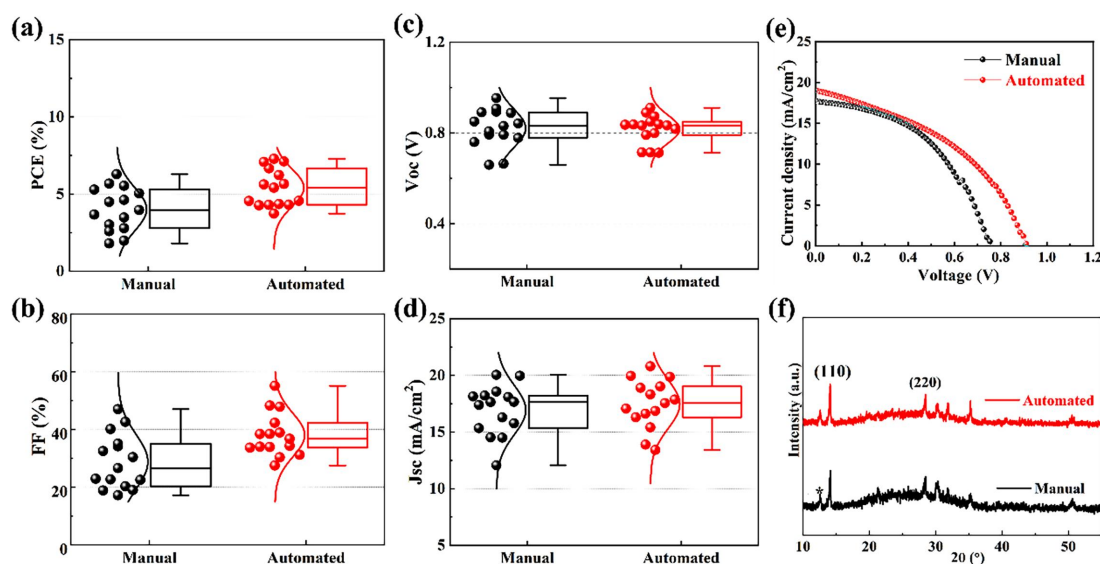


FIG. 2. (a) PCE distribution, (b) FF distribution, (c) V_{oc} distribution, (d) J_{sc} distribution, (e) J–V curves of PSCs, and (f) XRD patterns of MAPbI₃ films under automated and manual preparation.

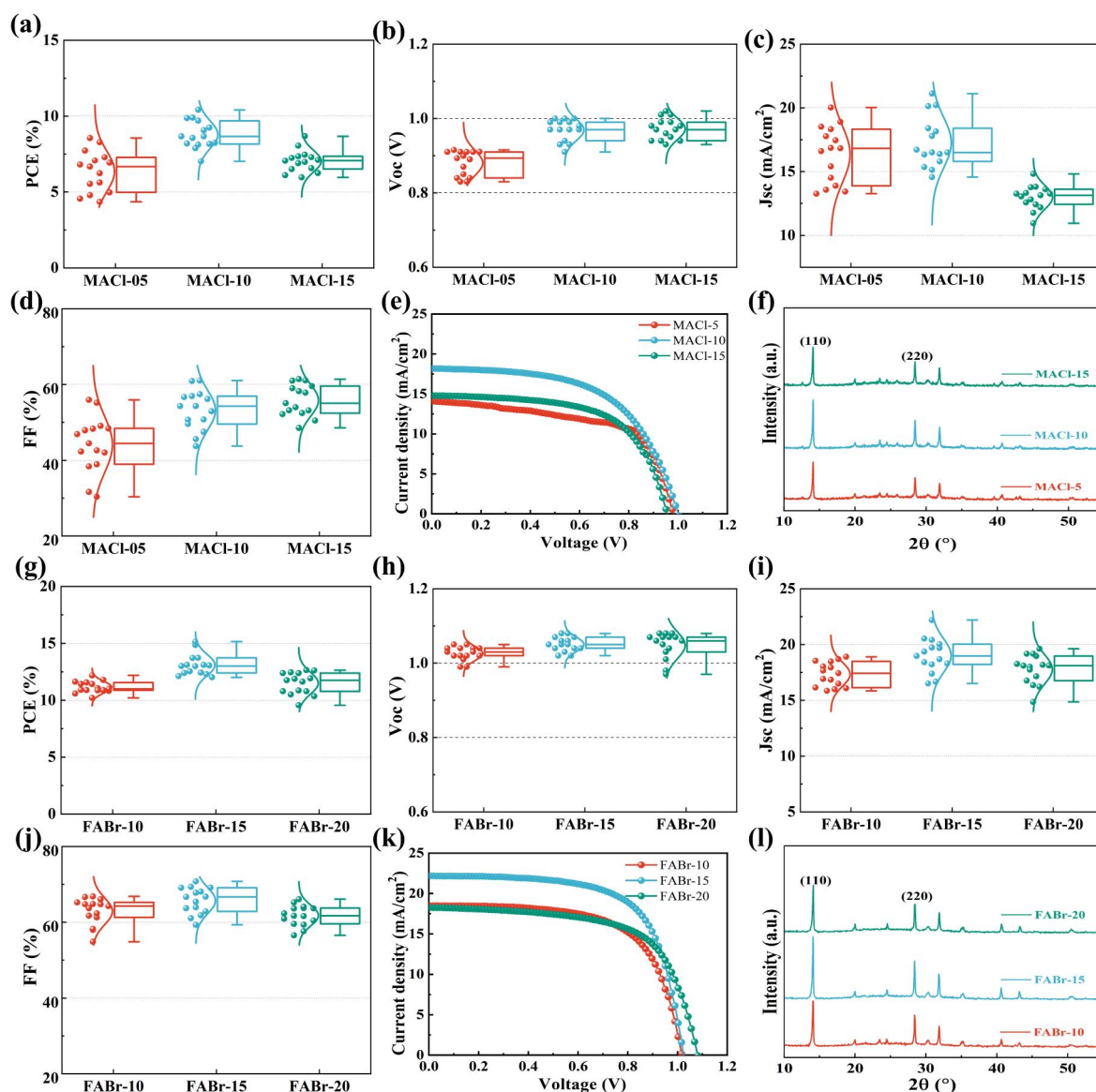


FIG. 3. (a) and (g) PCE distribution, (b) and (h) V_{oc} distribution, (c) and (i) J_{sc} distribution, (d) and (j) FF distribution, (e) and (k) J-V curves of PSCs, and (f) and (l) XRD patterns of the films fabricated by the automated preparation system at different additives.

to (110) and (220) crystal faces of MAPbI_3 . Additionally, distinct PbI_2 characteristic diffraction peaks were observed at $2\theta = 12.8^\circ$ (marked with “*” in the figure), indicating incompletely reacted PbI_2 in both films.

To further improve device performance and promote complete conversion of PbI_2 , MACl additive was introduced at concentrations of 5, 10, and 15 mg ml^{-1} (denoted MACl-5, MACl-10, and MACl-15). The PCE, FF, V_{oc} , and J_{sc} distribution and typical J-V curves under various MACl concentrations are shown in Figs. 3(a)–3(e), with detailed performance parameters summarized in Table S3. It can be seen that the MACl-10 devices exhibit the best PCE of 10.42%, V_{oc} of 1.00 V, J_{sc} of 18.15 mA cm^{-2} , and FF of 57.42%. The average PCE of

this MACl-10 group was $8.79\% \pm 0.91\%$, V_{oc} of 0.97 ± 0.03 V, J_{sc} of 17.27 ± 1.98 mA cm^{-2} , and FF of $53.23\% \pm 5.22\%$. In contrast, the MACl-5 group exhibited the lowest average performance, with a PCE of $6.36\% \pm 1.35\%$, largely due to a low FF of $44.17\% \pm 7.36\%$. For MACl-15 devices, the average PCE recovered to $7.03\% \pm 0.72\%$, but J_{sc} dropped significantly to 12.99 ± 0.93 mA cm^{-2} . The above results indicate that appropriate MACl addition helps improve the PSCs’ performance, while too low or too high addition amounts both lead to device performance degradation. Figure 3(f) shows XRD patterns of perovskite films under different MACl concentrations. All samples showed characteristic diffraction peaks of MAPbI_3 perovskite (110) and (220) crystal faces at $2\theta = 14.1^\circ$ and 28.5° , respectively.

MAcI-modified perovskite films showed no obvious PbI_2 characteristic peaks at $2\theta = 12.8^\circ$, indicating complete conversion of PbI_2 to the perovskite phase. Among them, MAcI-10 group diffraction peak intensity was significantly higher than MAcI-5 and MAcI-15 groups, indicating higher crystallinity with preferential growth along the (110) crystal face, consistent with experimental results showing optimal device performance for MAcI-10 PSCs. These results are because appropriate MAcI addition can promote perovskite film crystallinity and phase purity, directly reflected as PCE improvement.^{23,24}

Based on MAcI-10 optimization conditions, FABr was introduced at concentrations of 10, 15, and 20 mg ml^{-1} (denoted FABr-10, FABr-15, and FABr-20). FABr can simultaneously provide FA^+ organic cations and Br^- halogen anions, playing multiple synergistic roles during crystallization and film formation: FA^+ can partially substitute for MA^+ in the A-site, stabilizing the mixed-cation structure and suppressing non-radiative recombination,^{25,26} while Br^- participates in halide mixing, passivating defects at grain boundaries and surfaces, thereby reducing trap density and enhancing carrier lifetime and mobility.^{27,28} The photovoltaic parameters for these devices are summarized in Figs. 3(g)–3(k) and Table S4. The FABr-15 devices delivered the best performance, with a champion PCE of 15.15%, $V_{\text{oc}} = 1.02 \text{ V}$, $J_{\text{sc}} = 22.20 \text{ mA cm}^{-2}$, and FF = 66.75%. The average PCE for this group was $13.10\% \pm 0.93\%$, with corresponding $V_{\text{oc}} = 1.05 \pm 0.02 \text{ V}$, $J_{\text{sc}} = 19.01 \pm 1.52 \text{ mA cm}^{-2}$, and

FF = $65.73\% \pm 3.46\%$. The low standard deviation (0.93%) highlights excellent reproducibility. The FABr-15 devices also show improved stability over controls under N_2 and air atmospheres (Fig. S4), but the absolute lifetime is only a few days. Further stability improvement requires complementary strategies such as compositional engineering, interfacial passivation, and advanced encapsulation. The FABr-10 group showed moderate improvements, with an average PCE of $11.15\% \pm 0.51\%$, while the FABr-20 group exhibited slightly reduced performance, likely due to excessive Br^- incorporation. Control experiments using manual preparation with the same optimal additives (MAcI-10 and FABr-15) gave a champion PCE of 12.93% with a standard deviation of 1.18% (Table S5 and Fig. S5), compared to 15.15% (standard deviation = 0.93%) for the automated system, indicating that additives mainly boost efficiency while automation ensures better reproducibility. To further verify the statistical significance of the performance improvement, an independent two-tailed Welch's t-test was performed (Table S6), showing statistically significant differences between the automated and manually fabricated PSCs for each additive condition. These results confirm that the automated system not only increases average efficiency but also does so consistently across different additives. For comparison, a summary of recent representative electrodeposited PSCs is provided in Table S7, demonstrating that our optimized efficiency (15.15%) is comparable to the best reported values for electrodeposited MAPbI_3 PSCs. Figure 3(l)

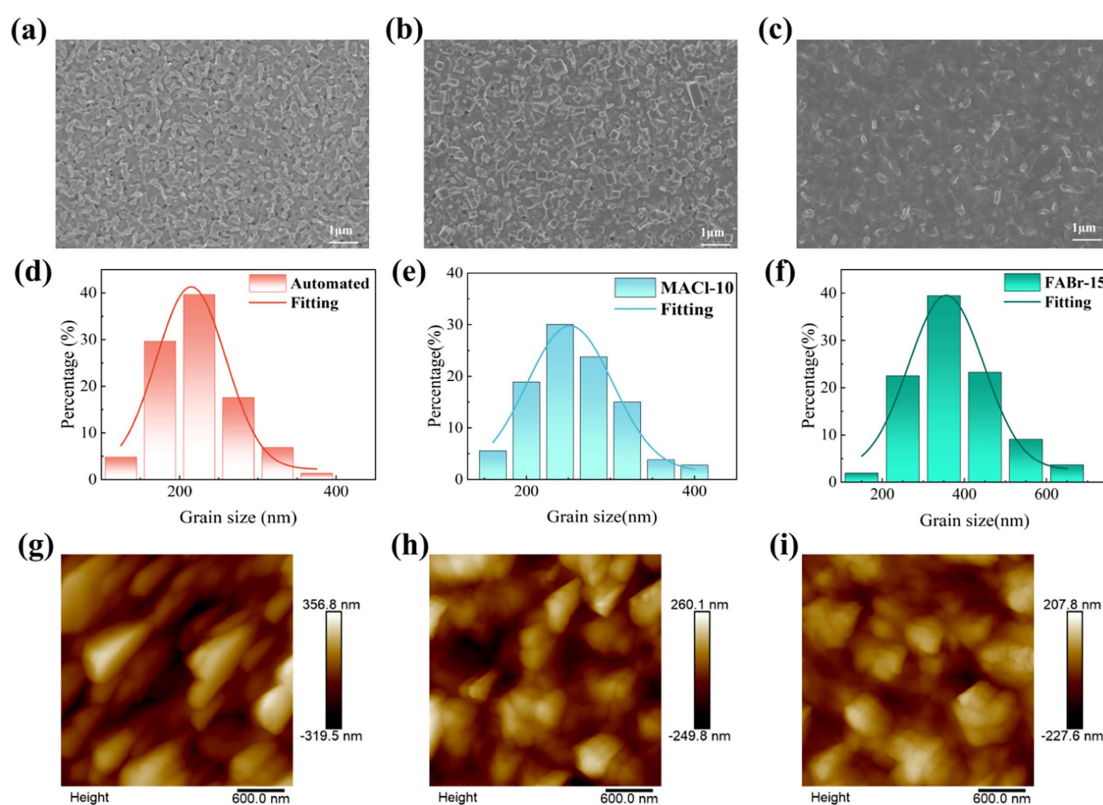


FIG. 4. SEM images, grain size statistic distributions, and AFM images of perovskite films fabricated with different additives: (a), (d), and (g) pristine MAPbI_3 ; (b), (e), and (h) MAcI-10; and (c), (f), and (i) FABr-15.

shows that the FABr-15 sample exhibits the strongest diffraction peaks, confirming the beneficial effect of optimal FABr addition on crystal quality.

To elucidate the underlying mechanisms of MACl and FABr on the automated electrodeposition prepared PSCs' performance, the surface morphology of perovskite films was characterized (Fig. 4). The SEM image of Fig. 4(a) exhibits a non-uniform morphology with small grains and noticeable pinholes for the pristine MAPbI₃ film. Upon addition of MACl-10, the film becomes more compact with larger grains and fewer defects [Fig. 4(b)]. The further addition of FABr-15 results in a dense, uniform film with large grains [Fig. 4(c)]. Grain size distributions of Figs. 4(d)–4(f) follow Gaussian profiles, with average grain sizes of 221 nm for the pristine perovskite MAPbI₃ film, 260 nm for the MACl-10 perovskite film, and 371 nm for the FABr-15 perovskite film, indicating that MACl and FABr significantly promote grain growth. The cross-sectional SEM images in Fig. S6 further confirm that the synergistic incorporation of MACl and FABr effectively improves film morphological uniformity. Compared with the pristine MAPbI₃ device without additives, the MACl and FABr optimized device exhibits clearer and more continuous interfaces between functional layers, significantly fewer pinholes within the perovskite layer, and markedly enhanced film compactness. AFM images of Figs. 4(g)–4(i) reveal that the root mean square (RMS) roughness decreases from 96.8 nm (pristine MAPbI₃ film) to 73.9 nm (MACl-10) and further to 50.2 nm (FABr-15), indicating that MACl and FABr synergistically reduce surface roughness. Collectively, such morphological optimization is beneficial for reducing interfacial recombination losses and facilitating efficient charge transport, consistent with the concurrent improvements in photovoltaic parameters observed in the PSCs.

The optical properties of the automated prepared perovskite films were investigated by UV-Vis absorption spectroscopy [Fig. 5(a)]. The MACl-10 and FABr-15 films exhibit effectively

enhanced absorption across the visible region compared with the pristine perovskite MAPbI₃ film, indicating improved light harvesting and photogenerated carrier density. Tauc plots in Fig. 5(b) yield optical bandgaps of 1.57 eV (pristine MAPbI₃ film), 1.58 eV (MACl-10), and 1.61 eV (FABr-15). The slight increase upon MACl addition is attributed to the transient role of Cl[−] during crystallization without incorporation into the lattice, while the further blue shift upon FABr addition arises from partial substitution of I[−] by Br[−], leading to lattice contraction and a wider bandgap conducive to higher V_{oc} .^{29–31} This interpretation is directly supported by XPS and EDS measurements (Figs. S7 and S8), which confirm the presence of Br in the FABr-15 film, and the decreased I 3d peak intensity is consistent with partial I[−]/Br[−] substitution. Figure 5(c) shows significantly enhanced PL intensity for the FABr-15 sample compared to the pristine and MACl-10 films. Time-resolved photoluminescence (TRPL) measurements (Fig. S9) reveal average carrier lifetimes of 1.9 ns (pristine), 2.7 ns (MACl-10), and 5.7 ns (FABr-15), indicating that the combined addition of MACl and FABr suppresses non-radiative recombination and reduces trap-state density more effectively than MACl alone. Electrochemical impedance spectroscopy (EIS) in Fig. 5(d) reveals a larger Nyquist semicircle radius for the MACl-10 and FABr-15 devices compared to the pristine device, indicating increased recombination resistance (R_{rec}) and reduced carrier recombination.^{32,33} Dark J–V curves in Fig. 5(e) show that the MACl-10 and FABr-15 devices exhibit a lower dark current density, consistent with reduced trap density and improved interfacial charge transport. To quantitatively assess non-radiative recombination, the light intensity dependence of V_{oc} was displayed in Fig. 5(f). Both devices exhibit a linear relationship between V_{oc} and the natural logarithm of light intensity, from which the ideality factor (n) was extracted using the diode equation³⁴

$$V_{oc} = \frac{nkT}{q} \ln \left(\frac{J_{sc}}{J_0} + 1 \right),$$

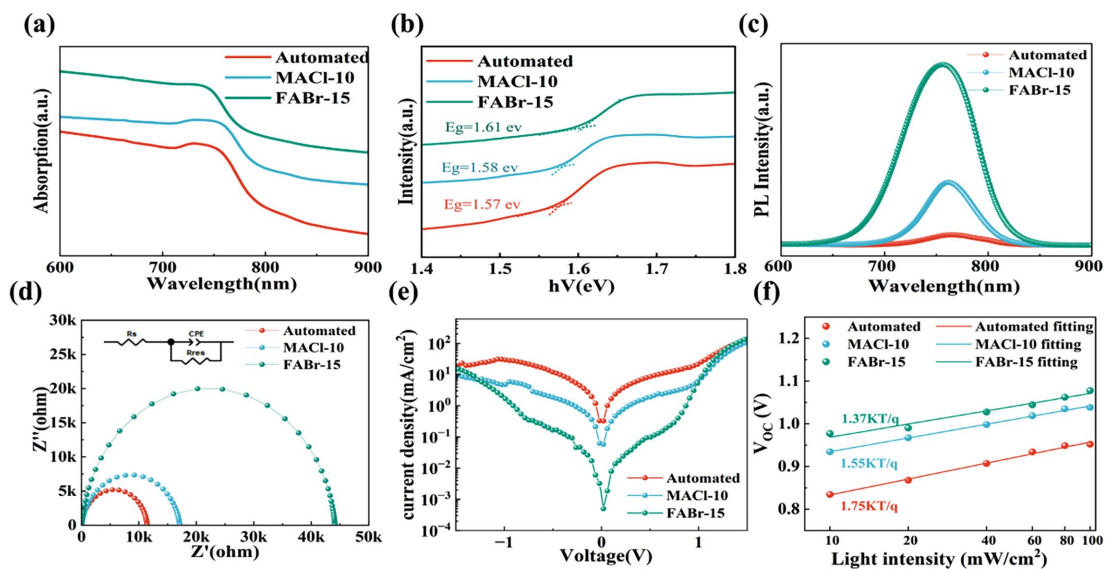


FIG. 5. (a) Absorbance spectra, (b) Tauc plot, and (c) PL of the perovskite films. (d) EIS spectra, (e) dark J–V curves, and (f) linear relationship of V_{oc} with the natural logarithm of light intensity of the PSCs.

where k is the Boltzmann constant, T is the temperature, q is the elementary charge, and J_0 is the dark saturation current density. The ideality factor decreases from 1.75 for the pristine device to 1.37 for the FABr-15 device, indicating a marked suppression of Shockley–Read–Hall recombination. Collectively, these results demonstrate that the MAcl and FABr additive combination synergistically enhances light absorption, reduces defects, and suppresses non-radiative recombination and leakage current, leading to an enhancement in device performance.

Beyond the results presented above, we discuss the prospects for large-area and tandem applications. Large-area devices are essential for the practical application of PSCs. Building on the automated system developed in this work, future research will systematically investigate the fabrication of large-area devices with controlled film uniformity, morphology consistency, and efficiency uniformity. Moreover, extending to Si-perovskite tandem solar cells offers a pathway beyond the single-junction efficiency limit.^{35,36} Although electrodeposition is inherently suitable for conformal coating on textured Si substrates,³⁷ the current efficiency of electrodeposited PSCs remains moderate, and inverted PSCs have rarely been reported. Hence, future work will also focus on electrodeposited inverted PSCs and their integration into Si-perovskite tandems.

In summary, this study addressed problems of rough process control and poor reproducibility in electrodeposited PSCs by designing and constructing a fully automated preparation system. Compared to manual preparation, the automated system prepared MAPbI₃ PSCs achieved a maximum PCE of 7.29%, a relative improvement of 16%, with concurrent improvements in both efficiency and reproducibility. Through MAcl and FABr additive optimization, the champion efficiency was further improved to 15.15%, with simultaneous improvements in V_{oc} , FF, and J_{sc} . Mechanism analysis showed that the synergistic action of MAcl and FABr suppresses non-radiative recombination and leakage current and improves light capture efficiency and carrier density, achieving comprehensive performance optimization.

See the [supplementary material](#) for the experimental section, automated system performance (Table S1), summaries of the photovoltaic parameters (Tables S2–S5), independent t-test statistical analysis (Table S6), design specification of equipment (Table S6), and electrodeposited PSCs' performance (Table S7); and additional figures showing the automated preparation control system (Fig. S1), device structure (Fig. S2), photograph of PSCs (Fig. S3), stability results (Fig. S4), photovoltaic parameters distribution (Fig. S5), cross-sectional SEM images (Fig. S6), XPS spectra (Fig. S7), EDS characterization (Fig. S8), and TRPL decay curves (Fig. S9).

The authors are thankful for the support of the National Natural Science Foundation of China (Grant No. 12204313).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tao Yu: Data curation (lead); Formal analysis (lead); Visualization (lead); Writing – original draft (equal). **Yuchong Chen:** Data curation

(equal); Formal analysis (equal); Investigation (equal). **Zhan Chen:** Data curation (equal); Formal analysis (equal); Visualization (equal). **Chenchao Feng:** Formal analysis (equal); Visualization (equal). **Diyun Xue:** Data curation (equal); Investigation (equal). **Centao Zhu:** Formal analysis (equal); Methodology (equal). **Chunhe Li:** Formal analysis (equal); Writing – original draft (equal). **Kuankuan Ren:** Formal analysis (lead); Funding acquisition (lead); Writing – original draft (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

REFERENCES

- A. Green Martin, D. Dunlop Ewan, M. Yoshita *et al.*, “Solar cell efficiency tables (version 67),” *Prog. Photovoltaics Res. Appl.* **34**(4), 482–496 (2026).
- Z. Chen, K. Ren, S. Yue *et al.*, “Performance-enhancing strategies for wide-bandgap perovskite solar cells and their tandem applications,” *ACS Energy Lett.* **10**(12), 6500–6548 (2025).
- J. H. Park and K. J. Y. Yoon Y S, “Fabrication processes for all-inorganic CsPbBr₃ perovskite solar cells,” *EcoMat* **5**(11), e12407 (2023).
- W. Chi and S. K. Banerjee, “Progress in the scale-up of single-junction and tandem perovskite solar devices,” *Chem. Eng. J.* **524**, 168953 (2025).
- S. Biberger, M. Spies, K. Schötz *et al.*, “Reactive spin coating based on real-time *in situ* feedback for improved control of perovskite thin film fabrication,” *J. Mater. Chem. C* **12**(18), 6415–6422 (2024).
- W. Wang, C. Zhou, Y. Zhou *et al.*, “Spin-coating free fabrication of α -FAPbI₃-based efficient inverted perovskite solar cells via mixed self-assembled molecules,” *Adv. Funct. Mater.* **36**, e23243 (2025).
- J. Liu, M. Zhang, X. Sun *et al.*, “Scalable fabrication of methylammonium-free wide-bandgap perovskite solar cells by blade coating in ambient air,” *Nano-Micro Lett.* **17**(1), 318 (2025).
- R. Pesch, A. Diercks, J. Petry *et al.*, “Hybrid two-step inkjet-printed perovskite solar cells,” *Sol. RRL* **8**(13), 2400165 (2024).
- Z. Zhang, Z. Li, Y. Chen *et al.*, “Progress on inkjet printing technique for perovskite films and their optoelectronic and optical applications,” *ACS Photonics* **10**(10), 3435–3450 (2023).
- X. Peng, J. Yuan, S. Shen *et al.*, “Perovskite and organic solar cells fabricated by inkjet printing: Progress and prospects,” *Adv. Funct. Mater.* **27**(41), 1703704 (2017).
- A. Khorasani, F. Mohamadkhani, M. Marandi *et al.*, “Opportunities, challenges, and strategies for scalable deposition of metal halide perovskite solar cells and modules,” *Adv. Energy Sustainability Res.* **5**(7), 2300275 (2024).
- Y. Gupta, M. Heydarian, M. Heydarian *et al.*, “Photostable inorganic perovskite absorber via thermal evaporation for monolithic perovskite/perovskite/silicon triple-junction solar cells,” *Prog. Photovoltaics Res. Appl.* **33**(7), 782–794 (2025).
- Y. Wang, Y. Gao, P. Liu *et al.*, “Scalable passivation of perovskite solar cells using evaporated CsPbCl₃,” *Sol. RRL* **9**(8), 2500042 (2025).
- X. Wu, X. Jin, Y. Yang *et al.*, “Insight into the defect chemistry and ion migration in perovskite fabricated by hybrid chemical vapor deposition,” *ACS Appl. Energy Mater.* **6**(24), 12198–12205 (2023).
- M. Al-Katirib, L. Perrin, L. Flandin *et al.*, “Electrodeposition in perovskite solar cells: A critical review, new insights, and promising paths to future industrial applications,” *Adv. Mater. Technol.* **8**(23), 2300964 (2023).
- Z. Gao, Y. Jiang, Y. Meng *et al.*, “A review of pinhole-free thin films based on electrodeposition technology: Theory, methods and progress,” *Molecules* **29**(23), 5615 (2024).
- X.-P. Cui, K.-J. Jiang, J.-H. Huang *et al.*, “Electrodeposition of PbO and its *in situ* conversion to CH₃NH₃PbI₃ for mesoscopic perovskite solar cells,” *Chem. Commun.* **51**(8), 1457–1460 (2015).
- H. Chen, Z. Wei, X. Zheng *et al.*, “A scalable electrodeposition route to the low-cost, versatile and controllable fabrication of perovskite solar cells,” *Nano Energy* **15**, 216–226 (2015).

- ¹⁹P. Lv, P. Zhang, Z. Chen *et al.*, “The preparation of all-inorganic CsPbI₂₋₃Br_{1+x} perovskite solar cells based on electrodeposited PbO₂ film,” *Sol. Energy* **207**, 618–625 (2020).
- ²⁰C. Li Q and L. I. C. Lu *et al.*, “Electrodeposition of lead dioxide induces the fabrication of perovskite FAPbI₃ film and electron-transport-layer-free solar cells,” *Sol. Energy* **233**, 515–522 (2022).
- ²¹Y. Xiang, T. Yu, D. Xue *et al.*, “Electrodeposition preparation of perovskite solar cells with an efficiency exceeding 19%,” *Small* **21**, 2502302 (2025).
- ²²D. Xue, C. Zhu, Z. Chen *et al.*, “Octylammonium iodide modification of electrodeposition-converted PbI₂ for fabricating efficient and stable perovskite solar cells,” *Appl. Phys. Lett.* **128**(5), 053902 (2026).
- ²³M. Kim, G.-H. Kim, T. K. Lee *et al.*, “Methylammonium chloride induces intermediate phase stabilization for efficient perovskite solar cells,” *Joule* **3**(9), 2179–2192 (2019).
- ²⁴C. Wang, B. He, M. Fu, Z. Su, L. Zhang, J. Zhang, B. Mie, and X. Gao, “Influence of MACl on the crystallization kinetics of perovskite via a two-step method,” *Crystals* **14**(5), 399 (2024).
- ²⁵O. Hakami, J. F. Rasheed, T. Zelai *et al.*, “Investigation of FAPbI₃ treatment’s potential for triple-cation multi-halide perovskite solar cells,” *Sol. Energy* **274**, 112590 (2024).
- ²⁶T. Yang, D. Huang, Z. Wang *et al.*, “Manipulating wide-band-gap perovskite compositions via Br sources for highly efficient perovskite/silicon tandem solar cells,” *ACS Appl. Energy Mater.* **7**(14), 5755–5764 (2024).
- ²⁷Y. Zhang, Q. Chen, H.-S. Yang *et al.*, “Water-repellent perovskites induced by a blend of organic halide salts for efficient and stable solar cells,” *ACS Appl. Mater. Interfaces* **13**(28), 33172–33181 (2021).
- ²⁸T. Zhang, H. Chen, Y. Bai *et al.*, “Understanding the relationship between ion migration and the anomalous hysteresis in high-efficiency perovskite solar cells: A fresh perspective from halide substitution,” *Nano Energy* **26**, 620–630 (2016).
- ²⁹C. Zhu, D. Xue, T. Yu *et al.*, “Electrodeposition of 11.05%-efficient inverted perovskite solar cells via MABr optimization,” *ACS Appl. Electron. Mater.* **8**, 688 (2026).
- ³⁰M. Pols, A. C. Van Duin, S. Calero *et al.*, “Mixing I and Br in inorganic perovskites: Atomistic insights from reactive molecular dynamics simulations,” *J. Phys. Chem. C* **128**(9), 4111–4118 (2024).
- ³¹Y. Xiang, R. Zheng, C. Li *et al.*, “Multistep optimization for the electrodeposited mixed perovskite FA_{1-y}Cs_yPbBr_xI_{3-x} solar cells,” *Nanotechnology* **35**(1), 015706 (2024).
- ³²R. Singh, M. Kundar, S. K. Pal *et al.*, “Bandgap engineered double-cation/double-halide quasi-cubic perovskite for highly efficient (>36%) indoor photovoltaics,” *IEEE J. Photovoltaics* **13**(6), 858–865 (2023).
- ³³S. Tombe, G. Adam, H. Heilbrunner *et al.*, “Optical and electronic properties of mixed halide (X= I, Cl, Br) methylammonium lead perovskite solar cells,” *J. Mater. Chem. C* **5**(7), 1714–1723 (2017).
- ³⁴L. Tang, L. Zeng, J. Luo *et al.*, “All-round passivation strategy yield flexible perovskite/CuInGaSe₂ tandem solar cells with efficiency exceeding 26.5%,” *Adv. Mater.* **36**(28), 2402480 (2024).
- ³⁵G. Y. Ayvazyan, L. A. Hakhoyan, H. R. Dashtoyan *et al.*, “Preparation and investigation of vacuum-deposited CH₃NH₃PbI_{3-x}Cl_x perovskite films on black silicon,” *J. Contemp. Phys. (Arm. Acad. Sci.)* **58**(1), 85–91 (2023).
- ³⁶S. Xia, F. Wang, P. Chen *et al.*, “Advanced sub-micron pyramid texture enables approximately 32% efficiency in perovskite/silicon tandem solar cells,” *Adv. Funct. Mater.* **36**, e25903 (2026).
- ³⁷Y. Wang, C. Gao, X. Wang *et al.*, “Controllable electrochemical deposition and theoretical understanding of conformal perovskite on textured silicon towards efficient perovskite/silicon tandem solar cells,” *J. Phys. Chem. C* **125**(5), 2875–2883 (2021).